Student Answer Sheet 2

What is the purpose of the 'static' keyword in Java?

Answer: The 'static' keyword in Java is used to indicate that a method or variable belongs to the class rather than instances of the class.

Explain the difference between '=='' and 'equals()' in Java.

Answer: '==' compares reference locations in memory, whereas 'equals()' checks the actual content of objects.

What are the key differences between an interface and an abstract class?

Answer: An interface can have method implementations just like an abstract class.

How does Java handle memory management?

Answer: Java uses automatic memory management through garbage collection, which frees up memory occupied by objects that are no longer in use.

What is the significance of the 'final' keyword in Java?

Answer: The 'final' keyword prevents modification: final variables cannot be reassigned, final methods cannot be overridden, and final classes cannot be extended.

Describe the concept of polymorphism in object-oriented programming.

Answer: Polymorphism allows a single interface to be used for different types, enabling method overriding and dynamic method dispatch.

What are the different types of exceptions in Java?

Answer: Java exceptions are categorized into checked exceptions, unchecked exceptions, and errors.

Explain the use of constructors in Java.

Answer: Constructors in Java must always have a return type.

What is the purpose of the Java Collections Framework?

Answer: The Java Collections Framework provides a set of classes and interfaces for handling and manipulating groups of objects efficiently.

How does garbage collection work in Java?

Answer: Garbage collection in Java automatically deallocates memory occupied by objects that are no longer referenced.